

BEC304 - NETWORK ANALYSIS

MODULE-3 TRANSIENT BEHAVIOUR & INITIAL CONDITIONS

- A N/W in which Branch current and node voltages are not changing with respect to time is said to be an steady state.
- The time interval required for the ckt from one condition to another by changing applied voltage or by changing branch currents & voltages from one value to a new value is called transition period.
- The response or o/p of N/W during transition period is called transient response of a N/W.
- If the N/W condition is not disturbed after transition period, then the N/W attain study state at infinite time is called study state response.
- The complete response of N/W is determined with the help of DE.
- Initial condition provide information regarding how elements behave during switching state.
- This knowledge is necessary in understanding non linear ckt.
- The initial condition is condition that exist in a N/W at the instant of switching operation.
- The reference time is $t=0$ one or more switches are operated.
- The condition just before switching operation will be represented as I^{0-} , V^{0-} condition just after

- switching will be represented as $I(t)$ & $V(t)$.
- Final cond can be obtained by allowing time t to be infinite of initial cond
 - in solving the problem we divide the t in following ways.

- just before switching ($t = -\infty$, $t = 0^-$)
- just after switching ($t = 0^+$)
- after switching ($t > 0$)

→ Procedure for evaluating initial condition.

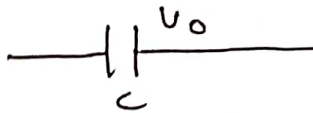
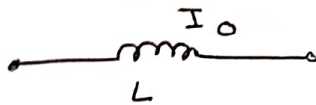
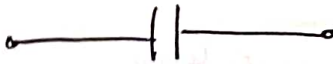
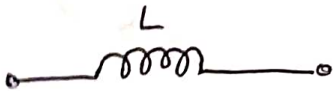
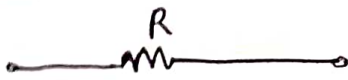
- Draw the eq n/w at $t = 0^-$ before switching action takes place
i.e. ($t = -\infty$, $t = 0^-$), the n/w is under steady state condition, hence find the current flowing through the inductor $i_L(0^-)$ & voltage across capacitor $V_C(0^-)$
- Draw the eq n/w at $t = 0^+$ immediately after switching replace all the inductor with o/pkt or with current sources $i_L(0^+)$ and replace all capacitor by short ckt's or voltage sources $V_C(0^+)$ resistor are kept as it is in the n/w.
- Initial voltage or currents in the n/w are determined from the equivalent n/w at $t = 0^+$
- Initial condition i.e. $\frac{di(0^+)}{dt}$, $\frac{dV(0^+)}{dt}$ and

$$\frac{d^2 i(0^+)}{dt^2} + \frac{d^2 V(0^+)}{dt^2}$$

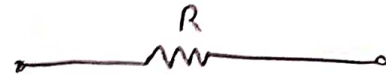
are determined by writing integro differential eqn at $t > 0$ i.e., after the switching action by making

Use of initial conditions

Element with Initial condition

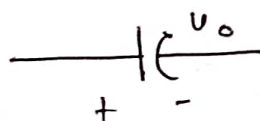
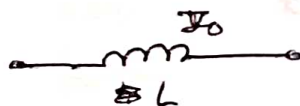
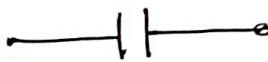
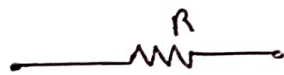


Equivalent ckt at $t = 0^+$

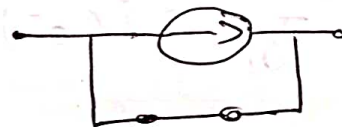


element with final condition at $t = \infty$

Element with Final condition



Equivalent ckt at $t = \infty$



Inductor

$$I = \frac{1}{L} \int v dt$$

$$v = L \frac{dI}{dt}$$

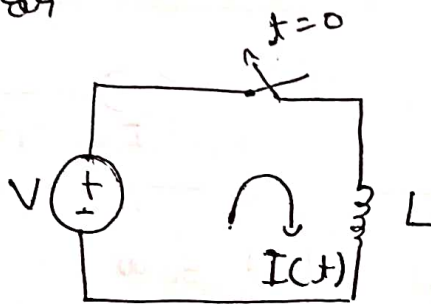
capacitor

$$I = C \frac{dv}{dt}$$

$$v = \frac{1}{C} \int I dt$$

• Initial condition and final condition.

1) Inductor



The current through L is

$$I_t = \frac{1}{L} \int_{-\infty}^t v dt$$

$$I_{(t)} = \frac{1}{L} \int_{-\infty}^{0^-} v dt + \frac{1}{L} \int_{0^-}^t v dt$$

$$I_{(t)} = i(0^-) + \frac{1}{L} \int_{0^-}^t v_{(t)} dt$$

Put $t=0^+$ on both sides

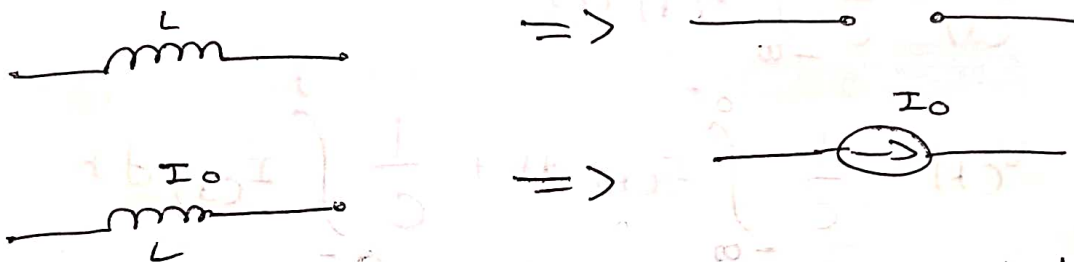
$$I_{(0^+)} = i(0^-) + \frac{1}{L} \int_{0^-}^{0^+} v(0^+) dt$$

$$\boxed{I(0^+) = I(0^-)}$$

from the above eqn it is clear that the current in an inductor cannot change instantaneously if $I(0^-) = 0$, $I(0^+) = 0$ this means at $t=0$ inductor act as an open ckt

If $I(0^-) = I_0$, $I(0^+) = I_0$, then the inductor can be ~~tauch~~ considered as current source of I_0 amps

$\therefore$ The equivalent ckt is



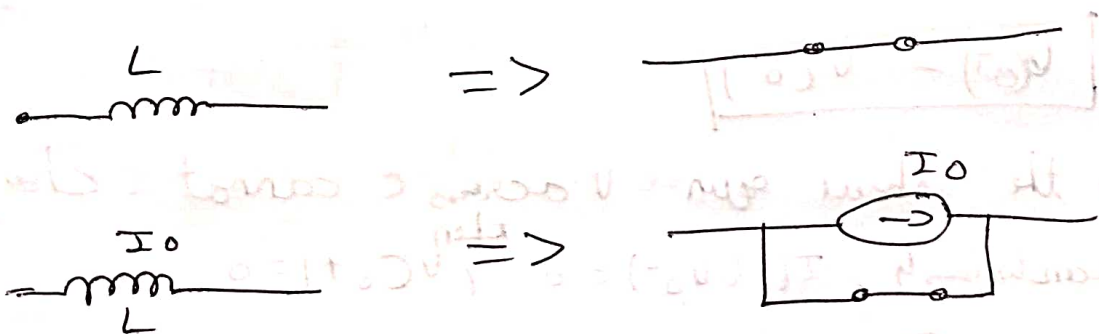
The final condition equivalent ckt of an inductor is derived from basic relationship.

$$V = L \frac{dI}{dt}$$

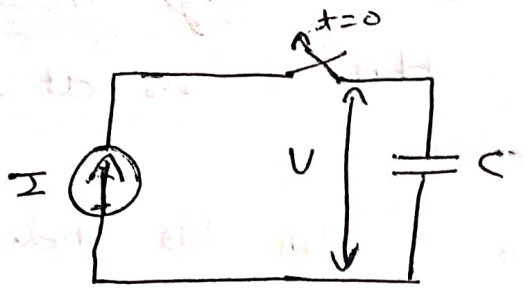
under steady state condition $\frac{dI}{dt} = 0$

$$\therefore V = 0$$

$\therefore L$ acts as short ckt at $t=0$



2) Capacitor



The V across C is given by

$$V_C(t) = \frac{1}{C} \int I_C(t) dt$$

$$V_C(t) = \frac{1}{C} \int_{-\infty}^t I_C(t) dt$$

$$V_C(t) = \frac{1}{C} \int_{-\infty}^{0^-} I_C(t) dt + \frac{1}{C} \int_{0^-}^t I_C(t) dt$$

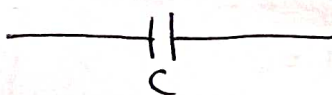
$$V_C(t) = V_C(0^-) + \frac{1}{C} \int_{0^-}^t I_C(t) dt$$

Put $t = 0^+$

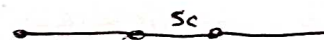
$$V_C(0^+) = V_C(0^-) + \frac{1}{C} \int_{0^-}^{0^+} I_C(t) dt$$

$$\boxed{V_C(0^+) = V_C(0^-)}$$

from the above eqn V across C cannot change instantaneously. If $(V_{0^-}) = 0$ then $(V_{0^+}) = 0$ at $t = 0^+$



$\Rightarrow$



If $V(0^-) = \frac{V_0}{C}$ then $V(0^+) = \frac{V_0}{C}$ then capacitor act as voltage source of V_0 volt



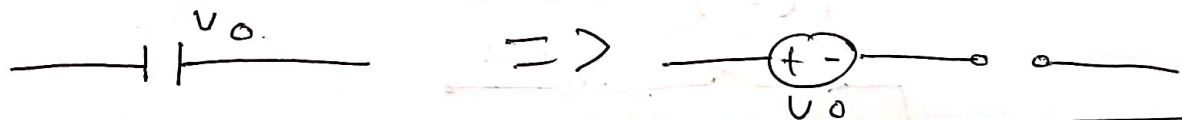
The final condition eq. ckt of a capacitor is derived from basic relationship

$$I = C \frac{dV}{dt}$$

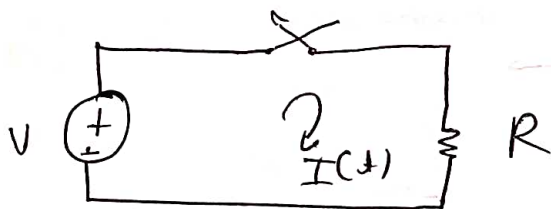
under steady state cond $\frac{dV}{dt} = 0$
 $\therefore$ at $t = \infty$

$$\boxed{\therefore I = 0}$$

i.e. at $t = \infty$ capacitor act as open ckt



3) Resistor :

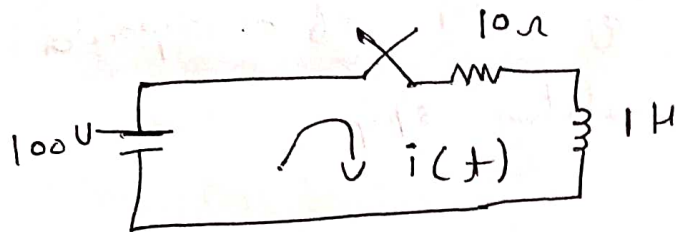


for R I & V are related by

$$V(t) = R I(t)$$

The current through R will change instantaneously if the V changes instantaneously.
 similarly V changes as current changes instantaneously.

• For the given n/w switch is closed at $t=0$ with 0 current in inductor find $i(t)$ and $\frac{d^2 i}{dt^2}$ at $t=0^+$



Case 1) at $t=0^-$, no current flows through the inductor $i(0^-) = 0 \text{ A}$

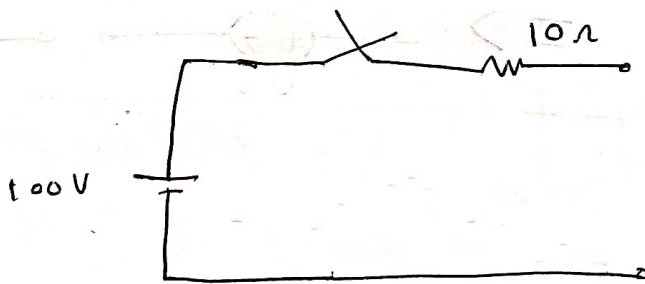
w.k.t

$$i(0^-) = i(0^+) = 0 \text{ A}$$

Case 2) at $t=0^+$

at $t=0^+$ the inductor acts as open ct

$$i(0^+) = i(0^-) = 0 \text{ A}$$



Case 3) at $t > 0$

for $t > 0$ L act as short ct.

apply KVL

$$100 - (10 i(t)) - L \frac{di(t)}{dt} = 0 \quad \dots \textcircled{1}$$

$$\frac{di}{dt} \Big|_{t=0^+} = 0$$

$$100 - 10 \frac{di(0^+)}{dt} - 1 \times \frac{d^2 i(0^+)}{dt^2}$$

$$100 = \frac{di(0^+)}{dt}$$

$$\frac{di(0^+)}{dt} = 100 \text{ A/s}$$

Diff eqn ① w.r.t to t

$$0 - 10 \times \frac{di(t)}{dt} - 1 \frac{d^2 i(t)}{dt^2} = 0$$

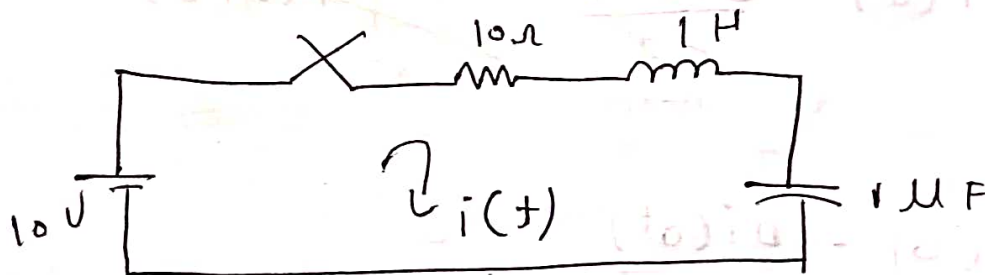
at $t = 0^+$

$$-10 \times \frac{di(0^+)}{dt} = 1 \frac{d^2 i(0^+)}{dt^2}$$

$$\frac{d^2 i(0^+)}{dt^2} = -10 \times 100$$

$$\frac{d^2 i(0^+)}{dt^2} = -1000 \text{ A/s}^2$$

2) For the n/w shown



case i at $t = 0^-$

for $t = 0^-$ the current flowing through L is 0 & across C = 0

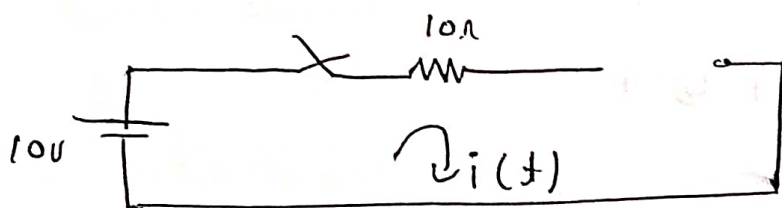
$$V_{(0^-)} = 0V$$

Case 2:

Case 2:
 $\omega^t = 0^t$

L will be open ckt

C will be short ckt



$$i(0^+) = 0 \text{ A}$$

$$V(0^+) = 0V$$

Case 3

at $t > 0$ (switch will be closed)

apply k u

$$10 - 10 i(t) - L \frac{di(t)}{dt} - \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = 0 \dots \textcircled{1}$$

$$dt = 0^+$$

$$10 - 10 i(\infty) - \frac{di(\infty)}{dt} - \frac{1}{s} \int_{-\infty}^{\infty} i(\infty) dt = 0$$

$$10 - 10(0) - \frac{di(0^+)}{dt} = 0$$

$$10 = \frac{di(0^+)}{dt}$$

$$\therefore \boxed{\frac{di(0^+)}{dt} = 10 \text{ A/s}}$$

diff eq wrt to t

$$-10 \frac{di(t)}{dt} - L \frac{d^2 i(t)}{dt^2} - \frac{1}{C} i(t) = 0$$

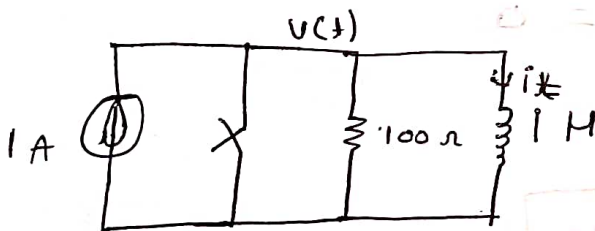
at $t = 0^+$

$$-10 \frac{di(0^+)}{dt} - 1 \frac{d^2 i(0^+)}{dt^2} - \frac{1}{1 \times 10^{-6}} i(0^+) = 0$$

$$-10 \times 10 = \frac{d^2 i(0^+)}{dt^2}$$

$$\boxed{\frac{d^2 i(0^+)}{dt^2} = -100 \text{ A/s}^2}$$

3) For the n/w shown in fig the switch is open at $t = 0$ calculate V , $\frac{dV}{dt}$, $\frac{d^2 V}{dt^2}$ at $t = 0^+$



soln: case (i) $t = 0^-$

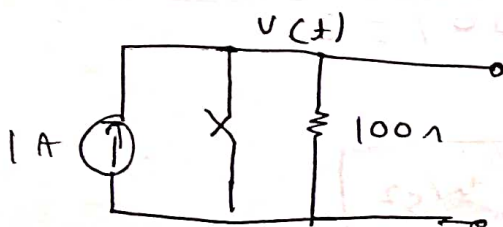
There is no current through Inductor.

$$\boxed{i(0^-) = 0 \text{ A}}$$

case (ii) at $t = 0^+$ (open ckt)

$$i(0^-) = 0 \text{ A}$$

$$\text{wkt } i(0^-) = 0 \text{ A} = i(0^+)$$



$$v(0^+) = 1 \times R$$

$$v(0^+) = 1 \times 100$$

$$\boxed{v(0^+) = 100 \text{ V}}$$

Case ③ $t > 0$ (switch will be closed)

apply KCL

$$\frac{v(t)}{100} + \frac{1}{L} \int v dt = 1 \dots \textcircled{1}$$

$$\frac{1}{100} \frac{dv(t)}{dt} + \frac{1}{L} v dt = 0 \dots \textcircled{2}$$

$$t = 0^+$$

$$\frac{1}{100} \frac{dv(0^+)}{dt} + \frac{1}{1} (100) = 0$$

$$\boxed{\frac{dv(0^+)}{dt} = -10000 \text{ V/s}}$$

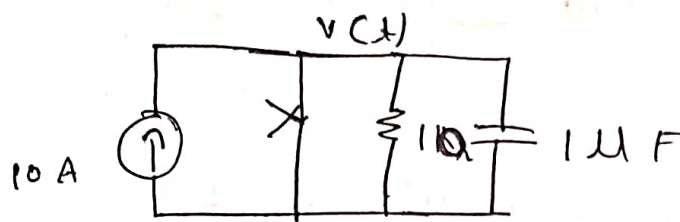
diff eqn ② with res to t

$$\frac{1}{100} \frac{d^2v(t)}{dt^2} + \frac{1}{L} \frac{dv(t)}{dt} = 0$$

$$\frac{1}{100} \frac{d^2v(t)}{dt^2} + \frac{1}{1} (-10000) = 0$$

$$\boxed{\frac{d^2v(0^+)}{dt^2} = 100000000 \text{ V/s}^2}$$

For the given circuit the switch is opened at $t=0$ solve for v , $\frac{dv}{dt}$, $\frac{d^2v}{dt^2}$ at $t=0^+$



case 1: ~~$t=0^-$~~ $t=0^-$

$$v(0^-) = 0$$

$$i(0^-) = 0$$

case 2: $t=0^+$

$$v(0^-) = v(0^+) = 0$$

$$i(0^-) = i(0^+) = 0$$

case 3: apply KCL

$$\frac{v(t)}{10} + C \frac{dv}{dt} = 10A \quad \dots \textcircled{1}$$

put $t=0^+$

$$\frac{v(0^+)}{10} + 1 \times 10^{-6} \frac{dv(0^+)}{dt} = 10$$

$$\boxed{\frac{dv(0^+)}{dt} = 10 \times 10^6 \text{ V/s}}$$

diff eqn $\textcircled{1}$ w.r.t to t

$$\frac{1}{10} \frac{dv(t)}{dt} + C \frac{d^2v}{dt^2} = 0$$

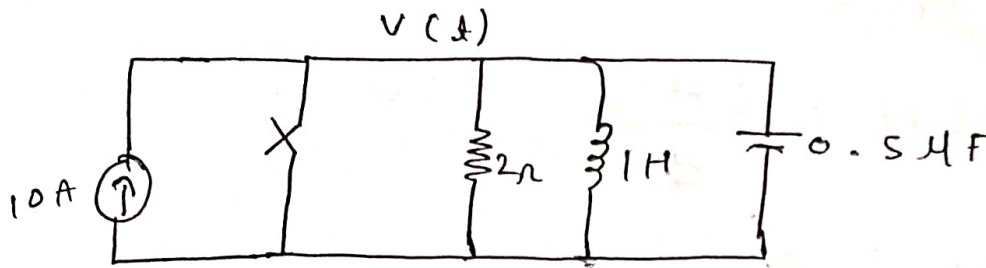
put $t = 0^+$

$$\frac{1}{10} \frac{dv(0^+)}{dt} + 1 \times 10^{-6} \frac{d^2v(0^+)}{dt^2} = 0$$

$$\frac{1}{10} (1 \times 10^6) + \frac{d^2v(0^+)}{dt^2} = 0$$

$$\frac{d^2v(0^+)}{dt^2} = -9.99 \times 10^{11} \text{ V/s}^2$$

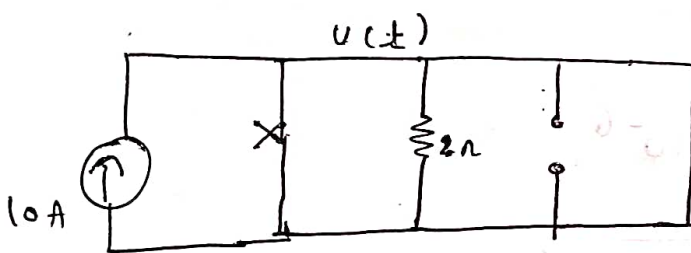
→ For the circuit shown below, find $v(t)$ at $t=0^-$
determine $v, \frac{dv}{dt}, \frac{d^2v}{dt^2}$



case ① $i(0^-) = 0$
 $v(0^-) = 0$

case ② at $t=0^+$

L will be open ckt
 C will be short ckt



$$i(0^+) = 0$$

$$v = i(0^+) \cdot R$$

$$v(0^+) = 0$$

$$v = 0 \times 20$$

$$\boxed{v = 0}$$

case ③ for $t > 0$

apply KCL

$$\frac{v(t)}{2} + \frac{1}{L} \int v(t) dt + C \frac{dv}{dt} = 10$$

$$\frac{v(t)}{2} + \frac{1}{4} \int v(t) dt + C \frac{dv}{dt} = 10$$

diff with resp to t

$$\frac{1}{2} \frac{dv}{dt} + \frac{1}{4} v(t) + C \frac{d^2v}{dt^2} = 10$$

put $t = 0^+$ in ①

$$\frac{v(0^+)}{2} + \frac{1}{4} \int_{-0}^{0^+} v(t) dt + 0.5 \mu \frac{dv(0^+)}{dt} = 10$$

$$\cancel{v(0)} \frac{0}{2} + 0.5 \times 10^{-6} \frac{dv(0^+)}{dt} = 10$$

$$\frac{dv(0^+)}{dt} = \frac{10}{0.5 \times 10^{-6}}$$

$$\boxed{\frac{dv(0^+)}{dt} = 20 \text{ MV/s}}$$

at $t = 0^+$ in ②

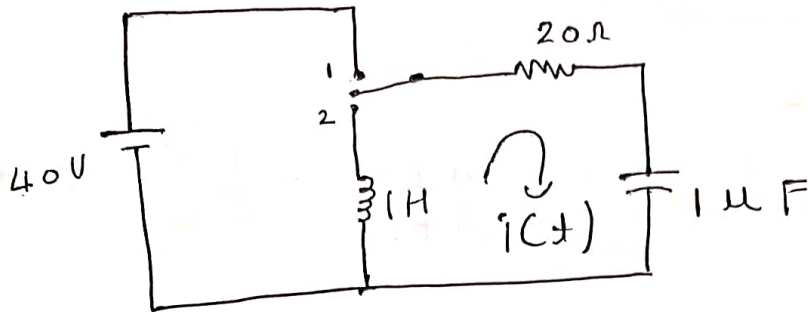
$$\frac{1}{2} \frac{dv(0^+)}{dt} + \frac{1}{4} \cancel{v(0^+)} + C \frac{d^2v}{dt^2} = 10$$

$$\frac{1}{2} (20 \text{ MV/s}) + 0.5 \times 10^{-6} \frac{d^2v}{dt^2} = 10$$

$$\frac{d^2v}{dt^2} = \frac{10 - 10 \text{ M}}{0.5 \times 10^{-6}}$$

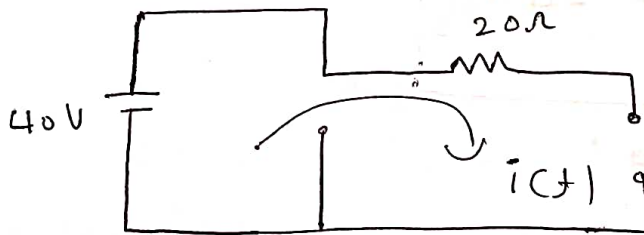
$$\boxed{\frac{d^2v}{dt^2} = -19.99 \times 10^{12} \text{ v/s}^2}$$

For the n/w shown the switch is changed from position ① to ② at $t=0$, study state cond having reached before switching find the value of i , $\frac{di}{dt}$, $\frac{d^2 i}{dt^2}$ at $t=0^+$



case 1 at $t=0^-$

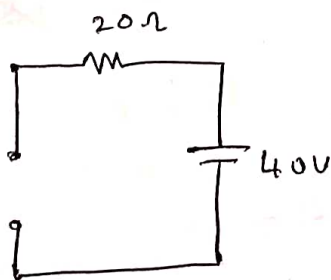
at $t=0^-$ the n/w attains steady state and hence L acts as a SC and capacitor as open ckt



$$i(0^-) = 0$$

$$V(0^-) = 40V$$

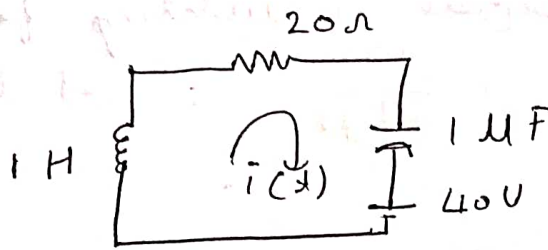
case ② at $(t=0^+)$ switch is changed from ① to ②



$$i(0^+) = 0$$

$$V(0^+) = 40V$$

iii) at $t > 0$ C switch B closed



apply KVL

$$\underline{40} - 20 i(t) - 1 L \frac{dI(t)}{dt} - \frac{1}{C} \int I(t) dt = 40$$

replace t by 0^+

$$0 - 1 \frac{dI(0^+)}{dt} - \cancel{\frac{1}{C} \int I(0^+) dt} = 40$$

$$\boxed{\frac{dI(0^+)}{dt} = -40 \text{ A/s}}$$

diff @ w.r.t to t

$$-20 \frac{dI(t)}{dt} - L \frac{d^2 I(t)}{dt^2} - \frac{1}{C} I(t) = 0$$

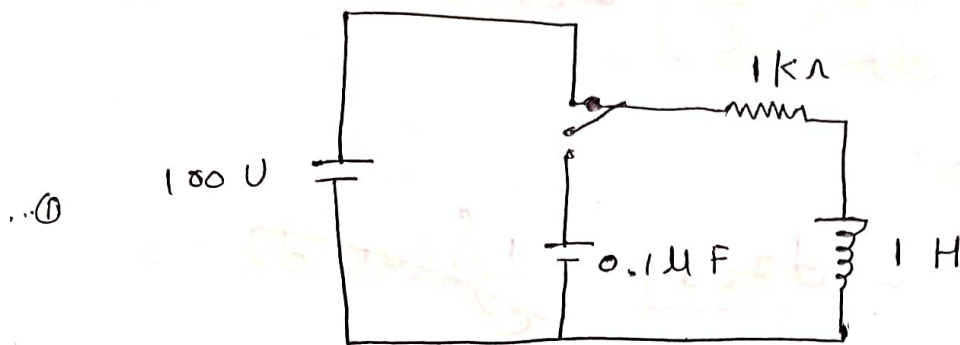
at $t = 0^+$

$$-1 \frac{d^2 I(0^+)}{dt^2} - 20(-40) - \cancel{\frac{1}{1 \times 10^{-6}} I(0^+)} = 0$$

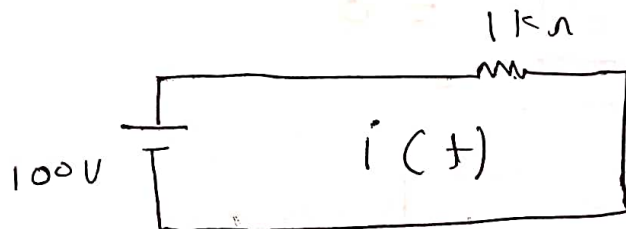
$$\therefore \frac{d^2 I(0^+)}{dt^2} = +800$$

$$\boxed{\frac{d^2 I(0^+)}{dt^2} = 800 \text{ A/s}^2}$$

→ For the n/w shown. switch is changed
 ② from position A to B at $t=0$ solve for
 i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t=0^+$

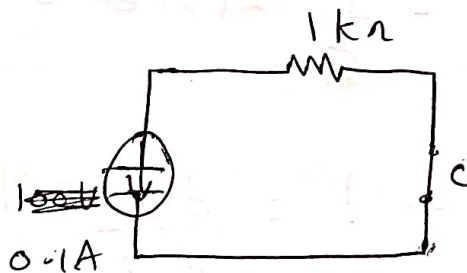


case 1: at $t=0^-$



$$i(0^-) = 0.1 \text{ A}$$

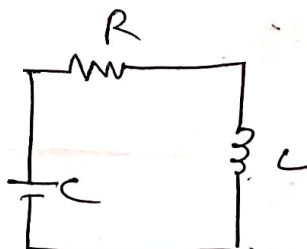
case 2: at $t=0^+$



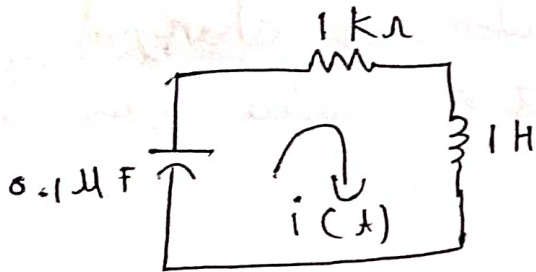
$$i(0^+) = 0.1 \text{ A}$$

$$V(0^+) = 100 \text{ V}$$

case 3:



$$i(0^+) = 0.1$$



$$-1K i(t) - L \frac{dI(t)}{dt} - \frac{1}{C} \int I dt = 0 \quad \dots \quad (1)$$

replace t as 0^+

$$-1 \times 10^3 i(0^+) - L \frac{dI(0^+)}{dt} - \frac{1}{C} \int I(0^+) dt = 0$$

$$-1000 - 1 \frac{dI(0^+)}{dt} = 0$$

$$\frac{dI(0^+)}{dt} = -1000 \text{ A/s}$$

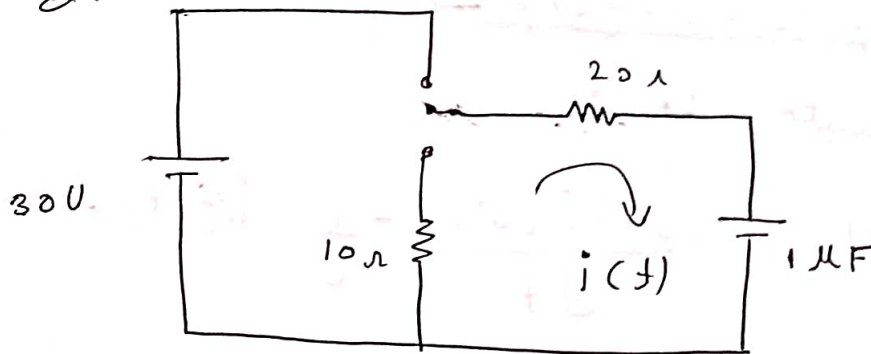
diff w.r.t t

$$-1K \frac{dI(0^+)}{dt} - L \frac{d^2 I(t)}{dt^2} - \frac{1}{C} I(t) \neq 0$$

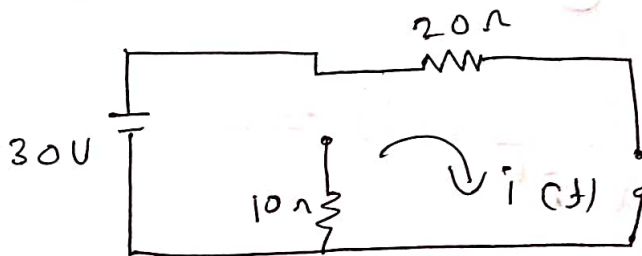
$$-1000(-1000) - \frac{1}{0.1 \times 10^{-6}} (0.1) = \frac{d^2 I(0^+)}{dt^2}$$

$$\frac{d^2 I(0^+)}{dt^2} = -9000 \times 10^3 \text{ A/s}^2$$

3) For the circuit shown switch present is closed from 1 to 2 at $t=0$ study and have reached before switching find the value of i , $\frac{di}{dt}$ at $t=0^+$



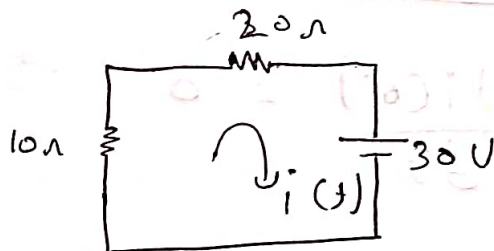
case i: $t=0^-$



$$i(0^-) = 0 \text{ A}$$

$$V(0^-) = 30 \text{ V}$$

case ii: $t=0^+$

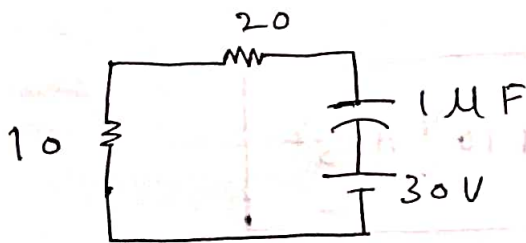


$$i(t) = -1$$

$$i(0^+) = -1 \text{ A}$$

$$\approx i(0^+) = 0$$

case iii: $t > 0$



apply KVL

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$$-10 i(t) - 20 i(t) - \frac{1}{C} \int i(t) dt = 30$$

diff @ with to t

$$-10 \frac{di(t)}{dt} - 20 \frac{di(t)}{dt} - \frac{1}{C} i(t) dt = 0$$

$$-30 \frac{di(t)}{dt} - \frac{1}{C} i(t) dt = 0 \dots (2)$$

$$-30 \frac{di}{dt} - \frac{1}{1 \times 10^{-6}} (-1) = 0$$

$$\frac{di}{dt} = \frac{1000000}{30}$$

$$\boxed{\frac{di}{dt} = +33.33 \times 10^3 \text{ A/s}}$$

$$-30 \frac{d^2 i}{dt^2} - \frac{1}{C} \frac{di(t)}{dt} = 0$$

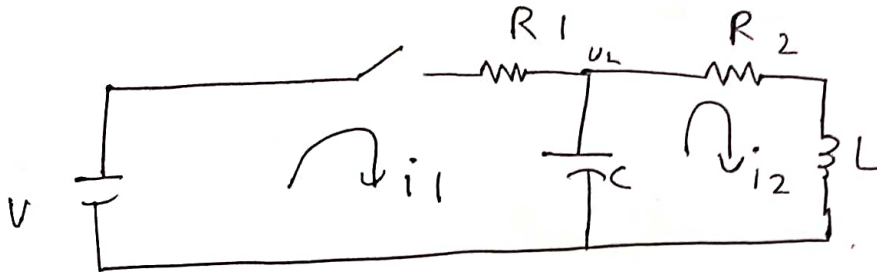
$$-30 \frac{d^2 i}{dt^2} - \frac{1}{1 \times 10^{-6}} (33.33 \times 10^3) = 0$$

$$-30 \frac{d^2 i(t)}{dt^2} = +3.33 \times 10^{10}$$

$$\boxed{\frac{d^2 i(t)}{dt^2} = -1.1 \times 10^9 \text{ A/s}^2}$$

-- For the ckt show all initial cond o
 bin $i_1(0^+)$, $i_2(0^+)$, $\frac{di_1}{dt}(0^+)$, $\frac{di_2}{dt}(0^+)$

$$\frac{d^2 i_1}{dt^2} \quad , \quad \frac{d^2 i_2}{dt^2}$$



at $T = 0^-$

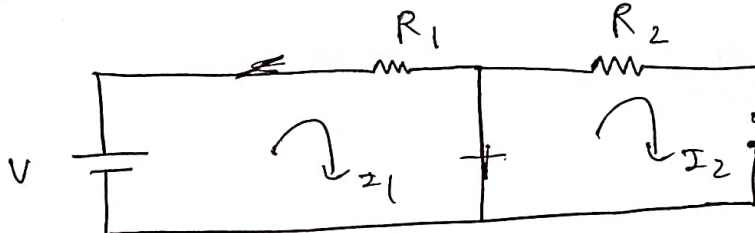
$$i_1(0^-) = 0$$

$$v_1(0^-) = 0$$

$$i_2(0^-) = 0$$

$$v_2(0^-) = 0$$

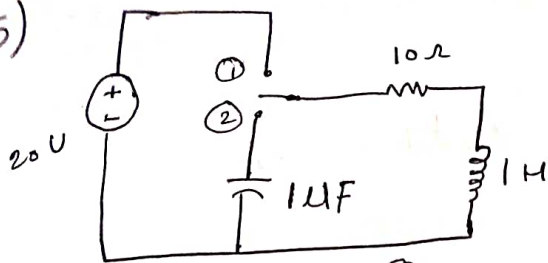
at $T = 0^+$



$$i_1(0^+) = \frac{V}{R_1}$$

$$i_2(0^+) = 0$$

5)



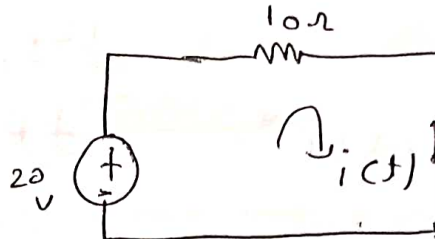
begin switched at ①

For a ckt shown the switch is changed from ① to ② at $t=0$, steady state have find the value of $\frac{di}{dt}$

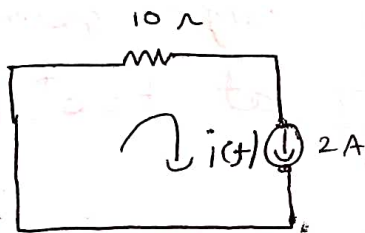
$$\frac{d^2 i}{dt^2} \text{ at } t=0$$

case i) $t = 0^-$

$$i(0^-) = \frac{20}{10} = 2A$$

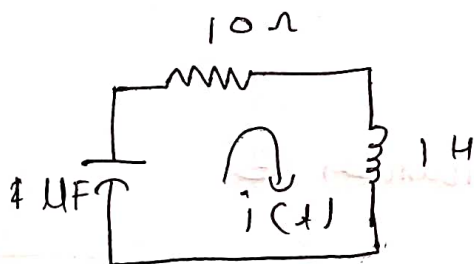


case ii) $t = 0^+$



$$i(0^+) = 2A$$

case iii)



$$-10 i(t) - L \frac{dI(t)}{dt} - \frac{1}{C} \int_{t=0^+}^t i(t) dt = 0$$

$$-10(2A) - 1 \frac{dI(0^+)}{dt} - \frac{1}{C} \int_0^0 2 dt = 0$$

$$\frac{dI}{dt} = -20 A/s$$

diff ① w.r.t to t

$$-10 \frac{d(i)t}{dt} - L \frac{d^2 I(t)}{dt^2} - \frac{1}{C} i(t) = 0$$

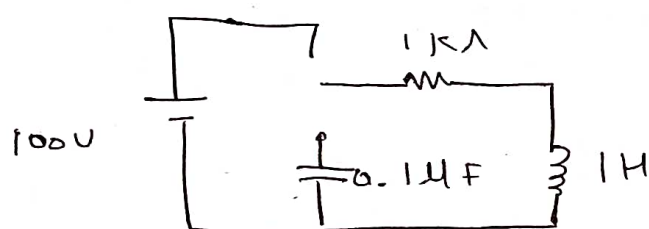
$$-10(-20) - \frac{1}{1 \times 10^{-6}} (2) = \frac{d^2 I(t)}{dt^2}$$

$$200 - \frac{2}{1 \times 10^{-6}} = \frac{d^2 I(t)}{dt^2}$$

$$\boxed{\frac{d^2 I(t)}{dt^2} = -1.9998 \times 10^6 \text{ A/s}^2}$$

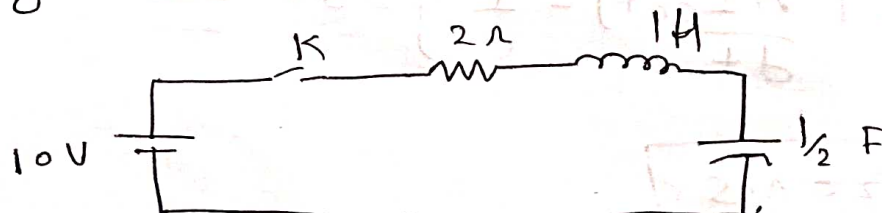
6) for the figure shown t is changed from a to b at $t=0$ solve, $\frac{di}{dt}$, $\frac{d^2 i}{dt^2}$ at $t=0^+$ $R=1000\Omega$

$L=1\text{H}$, $C=0.1\mu\text{F}$ and $V=100$ assume. C is initially uncharged



soln: same as before question number ②

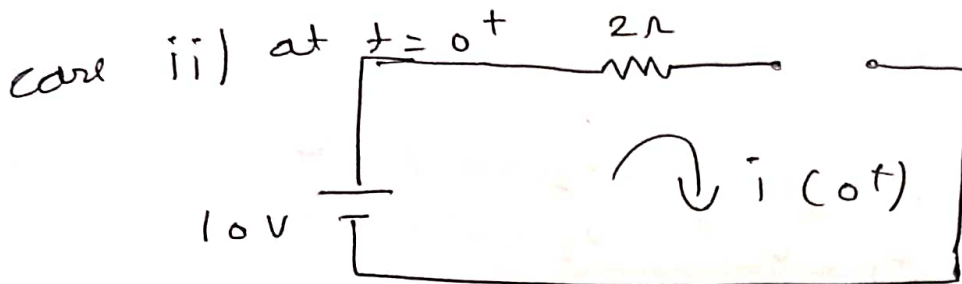
7) For the n/w shown switch K is closed at $t=0$ with C uncharged find $\frac{dI}{dt}$, $\frac{d^2 I}{dt^2}$ at $t=0^+$



case i) at $t = 0^-$

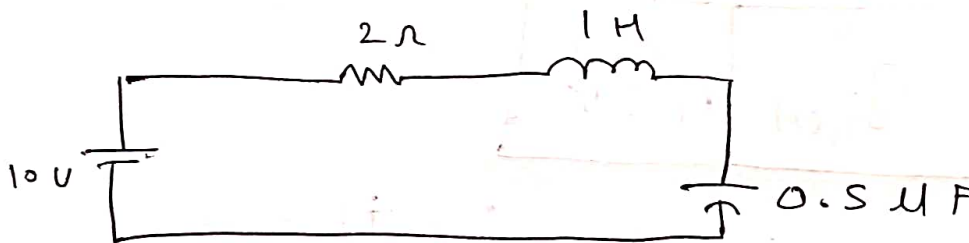
$$i(0^-) = 0$$

$$V(0^-) = 0$$



$$i(0^+) = 0$$

$$V(0^+) = 10V$$



$$+10 - 2i(t) - L \frac{di}{dt} - \frac{1}{C} \int i(t) dt = 0 \quad \dots \textcircled{1}$$

at $t = 0^+$

$$+10 - 2(i(0^+)) - L \frac{di(0^+)}{dt} - \frac{1}{0.5 \times 10^{-6}} \int i(t) dt = 0$$

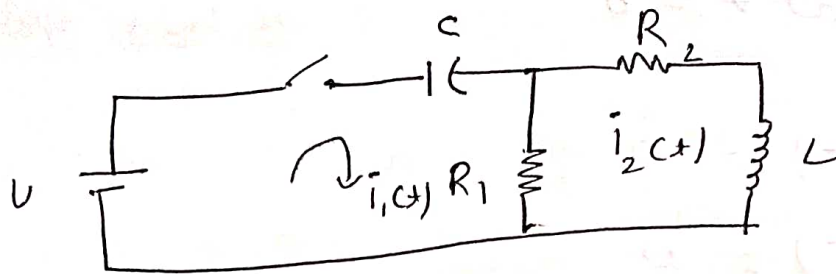
$$\frac{di(t)}{dt} = +10$$

can diff w.r.t t

$$\frac{d}{dt} \left(\frac{di(t)}{dt} \right) - 1 \frac{d^2 i}{dt^2} - \frac{1}{C} i(t) = 0$$

$$-(10)2 - \frac{1}{0.5 \times 10^{-6}} (0) = \left[\frac{d^2 i}{dt^2} = -20 A/s^2 \right]$$

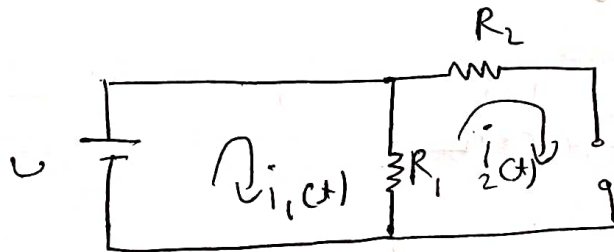
7)



case i: $t = 0^-$

$$\begin{aligned} i_1(0^-) &= 0 & i_2(0^-) &= 0 \\ v_1(0^-) &= V & v_2(0^-) &= V \end{aligned}$$

case ii: $t = 0^+$

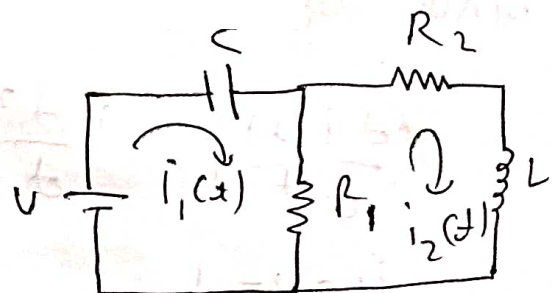


$$\begin{aligned} i_1(0^+) &= \frac{V}{R_1} & i_2(0^+) &= 0 \\ v_1(0^+) &= V & v_2(0^+) &= V \end{aligned}$$

case iii)

$$V - i_1(t)R_1 - \frac{1}{C} \int i_2(t) dt = 0 \quad \dots \quad (1)$$

put $t = 0^+$



$$\frac{dV}{dt} = \frac{di_1(t)}{dt} R_1 - \frac{di_2(t)}{dt} R_1 - \frac{1}{C} i(t)$$

$$= \frac{di_1(t)}{dt} R_1 - \left(R_1 + \frac{d}{dt} \right) \frac{1}{C} V = 0$$

$$= R_1 \left(\frac{di_1(t)}{dt} - \frac{di_2(t)}{dt} \right) - \frac{1}{C} V \quad (1)$$

$$-i_2(t) R_1 + i_1(t) R_1 - i_2(t) R_2 - L \frac{di_2(t)}{dt} = 0$$

$$\frac{di_2(t)}{dt} = \frac{-i_2(t) (R_1 + R_2) + i_1(t) R_1}{L}$$

$$\frac{dI_2(0^+)}{dt} = \frac{-\cancel{i_2(0^+) (R_1 + R_2)} + \frac{V}{R_1} (R_1)}{L}$$

$$\boxed{\frac{dI_2(0^+)}{dt} = +\frac{V}{L}}$$

from (2)

$$0 = \frac{1}{C} i(t) - R_1 \frac{di_1(t)}{dt} + R_1 \frac{di_2(t)}{dt} =$$

$$-\frac{1}{C} \frac{V}{R_1} - R_1 \frac{di_1(t)}{dt} + R_1 \left(+\frac{V}{L} \right) = 0$$

$$-\frac{V}{CR_1} = R_1 \left(\frac{di_1(t)}{dt} \right) - \frac{R_1 V}{L}$$

$$\left(-\frac{V}{CR_1^2} + \frac{V R_1}{L R_1} \right) \times \frac{1}{R_1} = \frac{di_1(t)}{dt}$$

$$\boxed{\left(-\frac{V}{CR_1^2} + \frac{V}{L} \right) = \frac{di_1(t)}{dt}}$$

$$-\frac{di_2(t)}{dt} R_1 + \frac{di_1(t)}{dt} R_2 - \frac{di_2(t)}{dt} R_1 - L \frac{d^2 I(t)}{dt^2} = 0$$

$$-\frac{di_2(t)}{dt} (R_1 + R_1) + \frac{di_1(t)}{dt} R_2 = L \frac{d^2 I(t)}{dt^2}$$

put $t = 0^+$

$$-\frac{di_2(0^+)}{dt} (R_1 + R_1) + \frac{di_1(0^+)}{dt} R_2 = L \frac{d^2 I(t)}{dt^2}$$

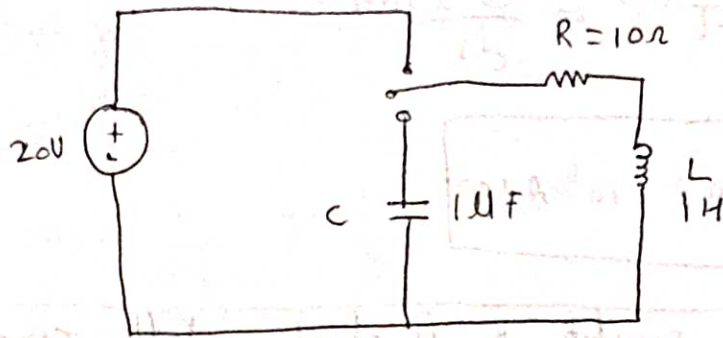
$$-\frac{V}{L} (R_1 + R_2) + \left(-\frac{V}{CR_1^2} + \frac{V}{L} \right) R_1 = L \frac{d^2 I(0^+)}{dt^2}$$

$$-\cancel{\frac{V R_1}{L}} - \frac{V R_2}{L} + \left(-\frac{V R_1}{C R_1^2} \right) + \cancel{\frac{V R_1}{L}} = L \frac{d^2 I(0^+)}{dt^2}$$

$$-\frac{V R_2}{L} - \frac{V R_1}{C R_1^2} = L \frac{d^2 I(0^+)}{dt^2}$$

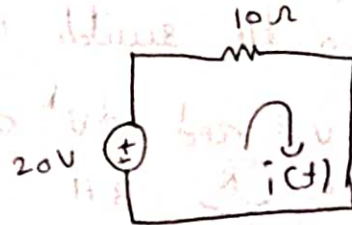
$$\boxed{\frac{d^2 I(0^+)}{dt^2} = -\frac{V R_2}{L^2} - \frac{V}{L C R_1}}$$

① For the circuit shown, the switch K is changed from position 2 at $t = 0$ steady state condition having been reached in position 1 find the value of i , $\frac{di}{dt}$, $\frac{d^2i}{dt^2}$ at $t = 0^+$

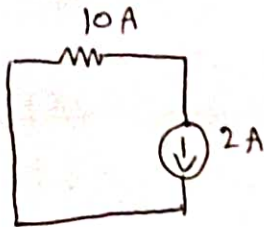


soln: case i) $t = 0^-$

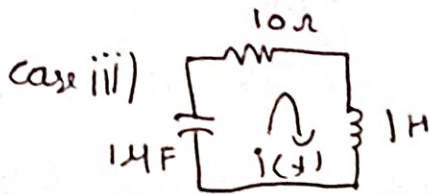
$$i(0^-) = \frac{20}{10} = 2A$$



case ii) $t = 0^+$



$$i(0^+) = 2A$$



$$-10i(t) - L \frac{di(t)}{dt} - \frac{1}{C} \int i(0^+) dt = 0$$

put $t = 0^+$

$$-10(2) - \frac{1}{1} \frac{di(0^+)}{dt} - \frac{1}{1} \int 2 dt = 0$$

$$\frac{di(0^+)}{dt} = -20 A/s$$

diff ① w a to $t = 0^-$

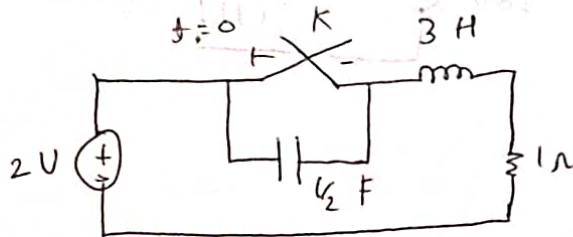
$$-10 \frac{d(i)}{dt} - L \frac{d^2 i(t)}{dt^2} - \frac{1}{C} i(t) = 0$$

put $t = 0^+$

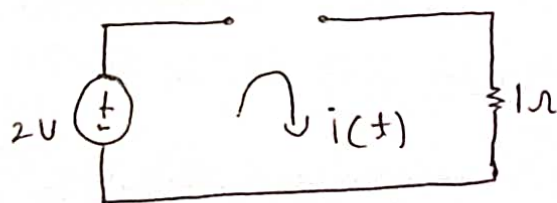
$$-10(-20) - \frac{1}{1 \times 10^{-6}}(2) = \frac{d^2 i}{dt^2}(0^+)$$

$$\therefore \frac{d^2 i(0^+)}{dt^2} = -1.9998 \times 10^6 \text{ A/s}^2$$

② For the ckt shown, the switch K is closed and the circuit is in steady state at $t=0$ the switch is opened. Determine the voltage across the switch V_K and dV_K at $t=0^+$



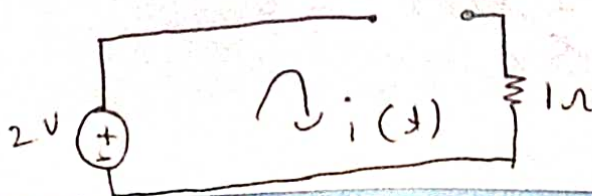
case i: $t = 0^-$



$$i(0^-) = 0$$

$$\therefore V_K(0^-) = 0$$

case ii: $t = 0^+$



$$i(0^+) = 0$$

$$\therefore V_k(0^+) = 0$$

case 3: $t > 0$

apply KVL

$$2 - C \frac{dV}{dt} - \frac{1}{L} \int V(t) dt - 1 dV(t) = 0 \quad \text{--- (1)}$$

put $t = 0^+$

$$2 - 0.5 \frac{dV}{dt} - \frac{1}{5} \int 0 dt - 0 = 0 \quad \text{--- (2)}$$

$$\frac{dV_k(0^+)}{dt} = \frac{2}{0.5}$$

$$\boxed{\frac{dV_k(0^+)}{dt} = 4 \text{ V/s}}$$

apply diff @ wrt t



$$0 - 0.5 \frac{d^2 V}{dt^2} - \frac{1}{L} (V(t)) = 0$$

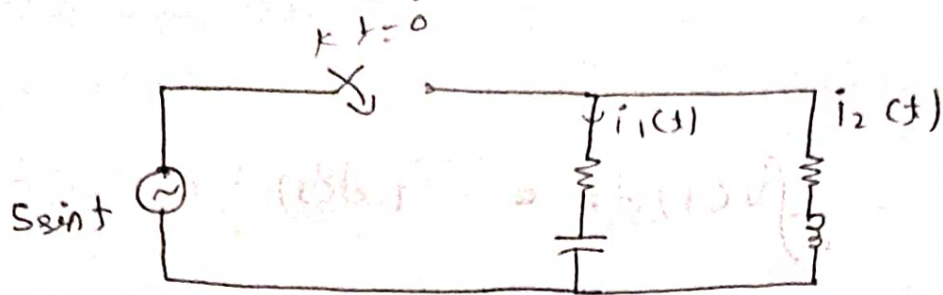
put $t = 0^+$

$$0 - 0.5 \frac{d^2 V(0^+)}{dt^2} - 0 = 0$$

$$\frac{d^2 V(0^+)}{dt^2} = \frac{0}{0.5}$$

$$\boxed{\therefore \frac{d^2 V(0^+)}{dt^2} = 0 \text{ V/s}^2}$$

3) In the ckt shown, the switch is closed at $t=0$ find $\frac{di_1(t)}{dt}$ and $\frac{di_2(t)}{dt}$



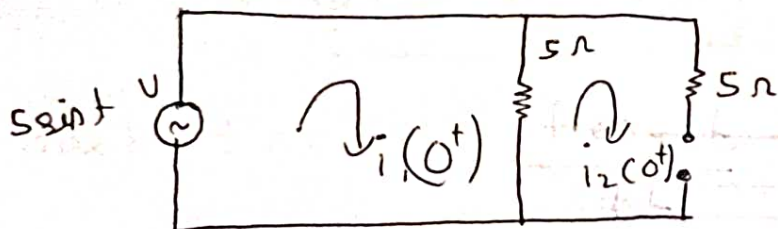
case i) case i $t=0^-$

The switch is open so no current flows

$$\therefore i_1(0^-) = 0$$

$$i_2(0^-) = 0$$

case ii : $t=0^+$



$$i_1(t) = \frac{V}{R} = \frac{5 \sin t}{5 \Omega}$$

$$i_1(t) = \frac{3.535}{5} = 0.7 \text{ A}$$

$$i_1(t) = 0.7 \text{ A}$$

$$i_2(t) = 0 \text{ A}$$

$$V = V_m \sin \omega t$$

$$V = 5 \sin t$$

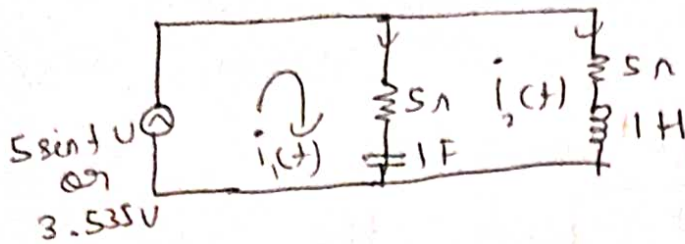
$$V_m = 5$$

$$V_{rms} = 0.707 \times V_m$$

$$V_{rms} = 0.707 \times 5$$

$$V_{rms} = 3.535$$

case 3: at $t > 0$
apply KVL



apply KVL to loop ①

$$3.535 - 5 i_1(t) + 5 i_2(t) - \frac{1}{C} \int i_1(t) dt + \frac{1}{C} \int i_2(t) dt = 0 \dots (1)$$

apply KVL to loop ②

$$-5 i_2(t) - L \frac{d i_2(t)}{dt} - \frac{1}{C} \int i_2(t) dt + \frac{1}{C} \int i_1(t) dt = 0$$

$$5 i_2(t) + 5 i_1(t) = 0$$

put $t = 0^+$

~~$$-5 i_2(0^+) - L \frac{d i_2(t)}{dt} - \frac{1}{C} \int i_2(0^+) dt + \frac{1}{C} \int i_1(0^+) dt = 0$$~~

~~$$5 i_2(0^+) + 5 i_1(0^+) = 0$$~~

$$\frac{5 i_1(0^+)}{1} = \frac{d i_2(t)}{dt}$$

$$\frac{5(0.7)}{1} = \boxed{\frac{d i_2(t)}{dt} = 3.5 \text{ A/s}}$$

can ① D.W. to +

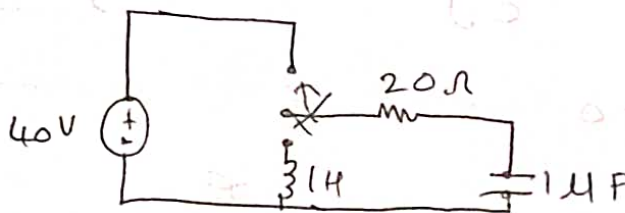
$$0 = 5 \frac{di_1(t^+)}{dt} + 5 \frac{di_2(t^+)}{dt} - \frac{1}{C} i_1(t^+)$$

$$+ \frac{1}{C} i_2(t^+) = 0$$

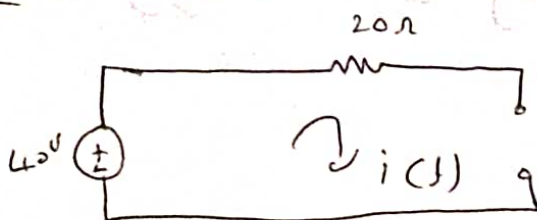
$$\frac{5(3.5) - 0.7}{5} = \frac{di_1(t^+)}{dt}$$

$$\boxed{\frac{di_1(t^+)}{dt} = 3.36 \text{ A/s}}$$

④. In the ckt shown, the switch K is moved from position 1 to position 2 at $t=0$, the steady state has been reached before switching - calculate $i(t)$, $\frac{di(t)}{dt}$, $\frac{d^2i(t)}{dt^2}$ at $t=0$

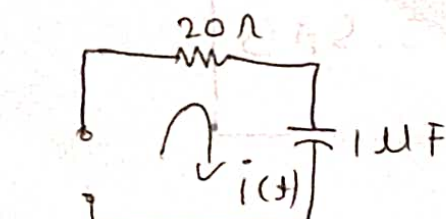


Case i: at $t=0^-$



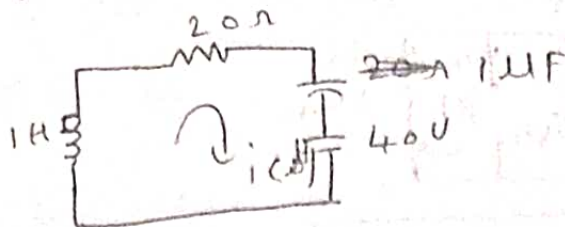
$$\boxed{i(0^-) = 0 \text{ A}}$$

Case ii) at $t=0^+$



$$\boxed{i(0^+) = 0 \text{ A}}$$

Case 3: $t > 0$



apply KVL

$$-20 i(t) - \frac{1}{C} \int_{0^-}^t I(x) dx - 40 - L \frac{dI(t)}{dt} = 0 \dots (1)$$

put $t = 0^+$

$$0 - \frac{1}{C} \int_{0^-}^{0^+} I(x) dx - 40 = \frac{dI(0^+)}{dt}$$

$$\therefore \frac{dI(0^+)}{dt} = -40 \text{ A/s}$$

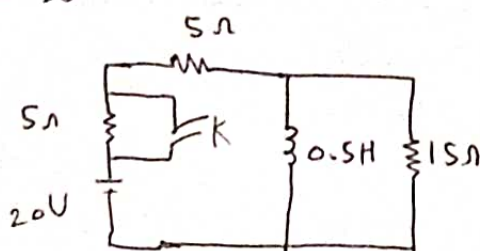
again diff wrt to t

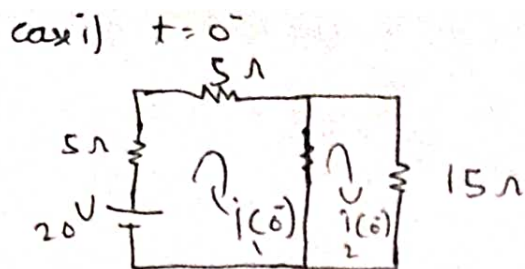
$$-20 \frac{dI(0^+)}{dt} - \frac{1}{C} I(0^+) - L \frac{d^2 I(t)}{dt^2} = 0$$

$$800 = \frac{d^2 I(t)}{dt^2}$$

$$\therefore \frac{d^2 I(t)}{dt^2} = 800 \text{ A/s}^2$$

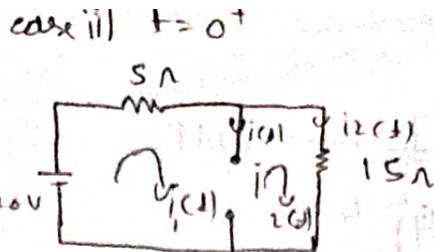
⑤ The switch K in the circuit is open for a long time at $t=0$ it is closed find $i_1(t)$, $i_2(t)$, $\frac{di_1(t)}{dt}$, $\frac{di_2(t)}{dt}$, $\frac{d^2 i_1(t)}{dt^2}$, $\frac{d^2 i_2(t)}{dt^2}$ at $t=0$





$$i_1(0^-) = \frac{20}{10} = 2 \text{ A}$$

$$i_2(0^-) = 0 \text{ A}$$

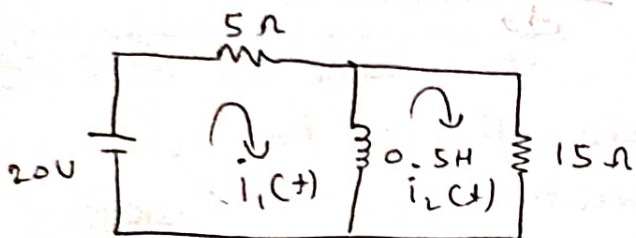


$$i_1(0^+) = 0 \text{ A}$$

$$i_2(0^+) = \frac{20}{20} = 1 \text{ A}$$

$$i_2(0^+) = 1 \text{ A}$$

case 3: $t > 0$



apply KVL ① loop

$$20 - 5i_1(t) - L \frac{di_1(t)}{dt} = 0$$

$$\frac{di_1(t)}{dt} = \frac{20}{0.5}$$

$$\frac{di_1(t)}{dt} = 40 \text{ A/s}$$

$$20 - 5 i_1(t) - 2 \frac{d i_1(t)}{dt} = 0 \dots \textcircled{1}$$

diff ① w.r to t

$$0 - 5 \frac{d i_1(t)}{dt} - 0.5 \frac{d^2 i_1(t)}{dt^2} = 0$$

$$\boxed{\frac{d^2 i_1(t)}{dt^2} = -400 \text{ A/s}^2}$$

apply KVL to loop ②

$$20 - 5 i_1(t) - 15 i_2(t) = 0 \dots \textcircled{2}$$

diff w.r to t

$$0 - 5 \frac{d i_1(t)}{dt} - 15 \frac{d i_2(t)}{dt} = 0$$

$$- \frac{5(40)}{15} = \frac{d i_2(t)}{dt}$$

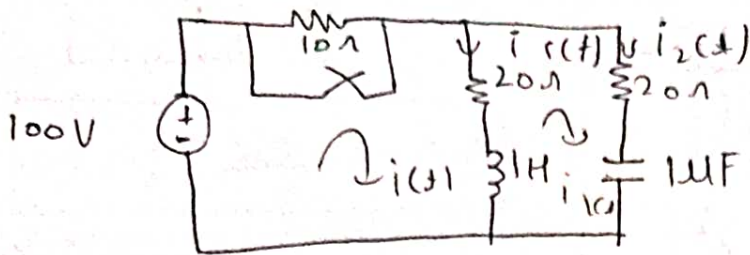
$$\boxed{\frac{d i_2(t)}{dt} = -13.33 \text{ A/s}}$$

eq ② D.W w.r to t

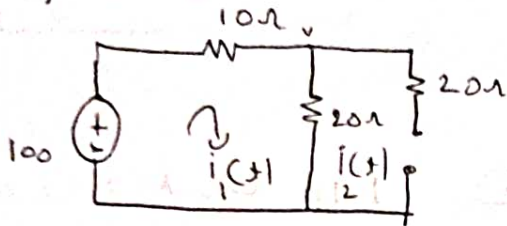
$$-5 \frac{d^2 i_1(t)}{dt^2} - 15 \frac{d^2 i_2(t)}{dt^2} = 0$$

$$\boxed{\frac{d^2 i_2(t)}{dt^2} = 133.33 \text{ A/s}^2}$$

b) In the ckt the switch across 10Ω is closed at $t=0$
 current $i_1(t)$, $i_2(t)$, $\frac{di_1(t)}{dt}$, $\frac{di_2(t)}{dt}$ at $t=0^+$



case i) $t=0^-$

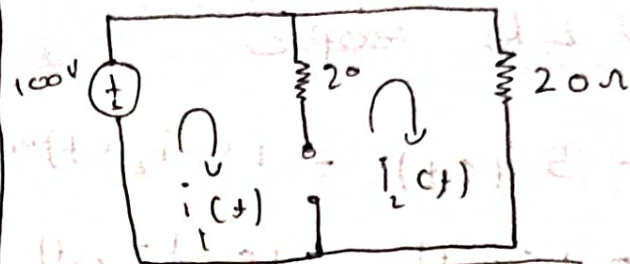


$$i_1(0^-) = \frac{100}{3\Omega} = \frac{10}{3} \text{ A}$$

$$i_2(0^-) = 0 \text{ A}$$

$$V_1(0^-) = I \times R = 66.66$$

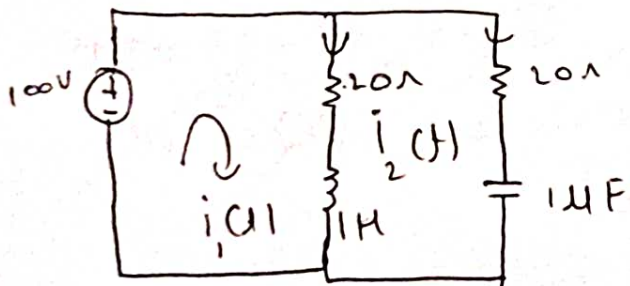
case ii) at $t=0^+$



$$i_1(0^+) = i_1(0^-) = \frac{10}{3}$$

$$i_2(0^+) = i_2(0^+) = 0 \text{ A}$$

case iii) $t \geq 0$



apply KCL

$$100 = 20i_1 + L \frac{di_1}{dt} \dots \textcircled{1}$$

$$100 = 20i_2 + \frac{1}{C} \int i_2 dt \dots \textcircled{2}$$

put $t=0^+$

$$100 = 20 \left(\frac{10}{3} \right) + L \frac{di_2(t)}{dt}$$

$$\frac{di_2(t)}{dt} = 100 - 66.66 = 33.33 \text{ A/s}$$

$$100 = 20 i_2(t) + \frac{1}{C} \int_{-\infty}^t i_2(t) dt$$

diff wrt t

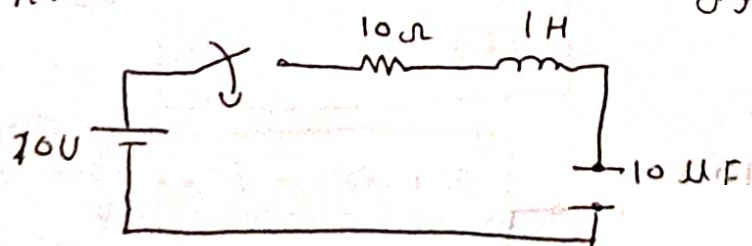
$$0 = 20 \frac{di_2(t)}{dt} + \frac{1}{C} i_2(t)$$

$$0 = 20 \frac{di_2(t)}{dt} + \frac{1}{1 \times 10^{-6}} \left(\frac{5}{3} \right)$$

$$-\frac{1666666.667}{20} = \frac{di_2(t)}{dt}$$

$$-8.3333 \times 10^4 \text{ A/s}$$

⑦ In the n/w shown find i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t=0^+$



case i at $t=0^-$

$$i(0^-) = 0 \text{ A}$$

case ii at $t=0^+$

$$i(0^-) = i(0^+) = 0 \text{ A}$$

apply KVL

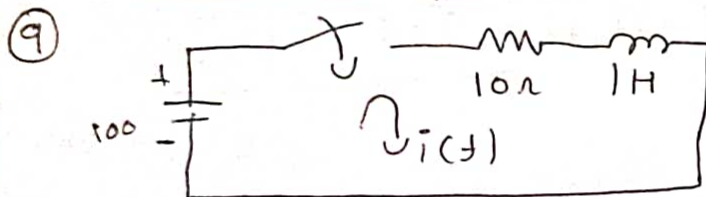
$$10 - 10i(\omega t) - L \frac{dI}{dt} - \frac{1}{C} \int_{-\infty}^{\omega t} I(\omega t) dt = 0 \quad \text{--- (1)}$$

$$10A = \frac{dI}{dt}$$

① P. w. to t

$$0 - 10 \frac{dI}{dt} - L \frac{d^2 I}{dt^2} - \frac{1}{C} I(\omega t) dt = 0$$

$$\frac{d^2 I}{dt^2} = -10(10) = -100 \text{ A/s}^2$$

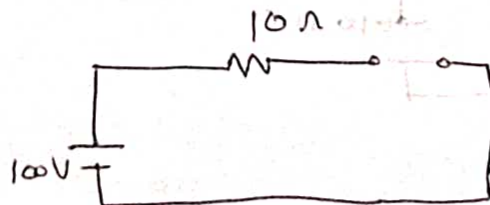


soln: $t = 0^-$

$$i(0^-) = 0A$$

case ii $t = 0^+$

$$i(0^+) = 0A$$



case iii $t > 0$

apply KVL

$$100 - 10i(\omega t) - L \frac{dI(\omega t)}{dt} = 0 \quad \text{--- (1)}$$

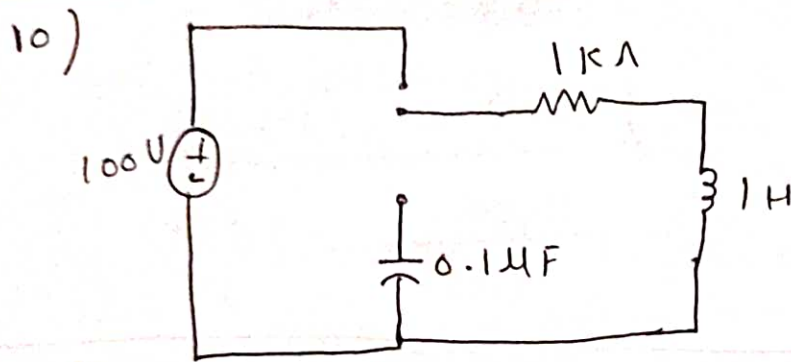
$$\boxed{\frac{di(t)}{dt} = 100 \text{ A/s}}$$

Diff @ $t=0^-$ to $t=0^+$

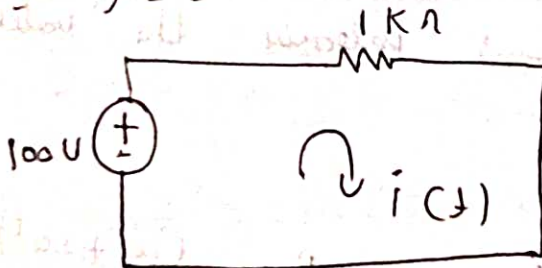
$$0 - 10 \frac{di(t)}{dt} = L \frac{d^2 i(t)}{dt^2}$$

$$\frac{L d^2 i(t)}{dt^2} = -10(100)$$

$$\boxed{\frac{d^2 i(t)}{dt^2} = -1000 \text{ A/s}^2}$$



case i: $t = 0^-$

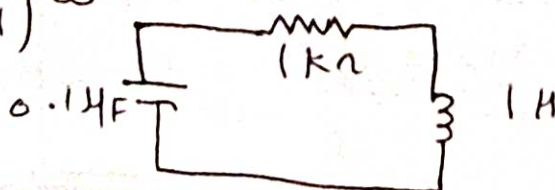


$$\boxed{i(0^-) = 0.1 \text{ A}}$$

case ii: $t = 0^+$

$$\boxed{i(0^-) = i(0^+) = 0.1 \text{ A}}$$

case (iii) at $t > 0$



$$-1000 (i(t)) - L \frac{di(t)}{dt} - \frac{1}{C} \int i dt = 0$$

$$\boxed{\frac{di(t)}{dt} = -100 A/s}$$

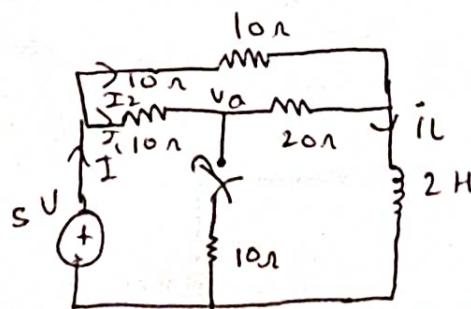
Diff 0 w.r.t to t

$$-1K \frac{di(t)}{dt} - L \frac{d^2 i(t)}{dt^2} - \frac{1}{C} i(t) = 0$$

$$-1000 (-100) - \frac{1}{0.1 \times 10^{-6}} (0.1) = \frac{d^2 i(t)}{dt^2}$$

$$\boxed{\frac{d^2 i(t)}{dt^2} = -900 \times 10^3 A/s^2}$$

12) In the circuit shown a steady state is reached with the switch open. At $t=0$ the switch is closed. Determine the value of $v_a(t)$ and $v_a(0^-)$



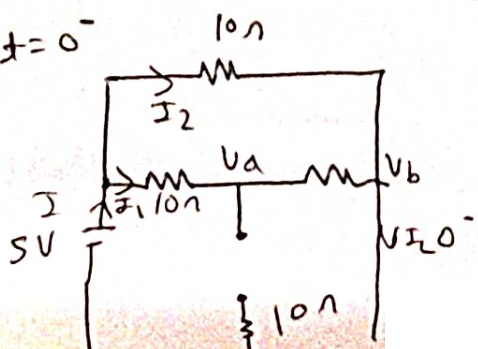
$$R_{eq} = (10 + 20) \parallel 10$$

$$R_{eq} = \frac{30 \times 10}{30 + 10} = 7.5 \Omega$$

$$I = \frac{V}{R} \quad I = \frac{5}{7.5}$$

$$\boxed{I = 0.666 A}$$

case II $t=0^-$



$$I_2 = I - I_1$$

$$I_2 =$$

$$\therefore I_1 = \frac{I \times 10}{10 + 10 + 20}$$

$$I_1 = \frac{0.666 \times 10}{40}$$

$$I_1 = 0.1666 \text{ A}$$

$$I_2 = 0.666 - 0.1666$$

$$I_2 = 0.5 \text{ A}$$

$$\therefore I \text{ at } t = 0^-$$

$$I(0^-) = 0.666$$

WKT

$$I(0^-) = I(0^+) = 0.666 \text{ A}$$

$$V_a = \frac{10 \times 5}{10 + 10 + 20}$$

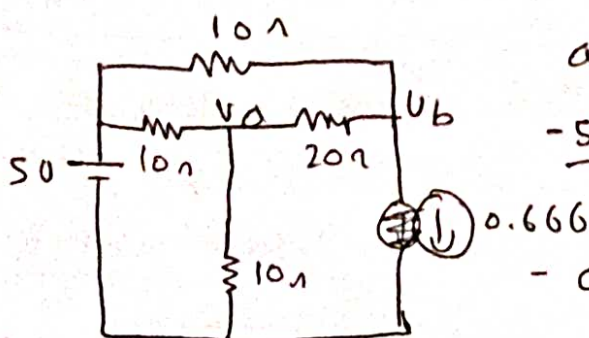
$$V_a =$$

$$I_1 = \frac{5 - V_a}{10}$$

$$V_a = 5 - 10(I_1)$$

$$V_a(0^-) = 3.334 \text{ V}$$

case iii at $t > 0$ switch is closed:



apply KCL at V_a & V_b

$$- \frac{5 + V_a(0^+)}{10} + \frac{V_a(0^+)}{10} + \frac{V_a - V_b}{20}$$

$$- 0.5 - 0.05V_b + 0.25V_a(0^+) = 0$$

$$0.25 V_a(0^+) - 0.05 V_b(0^+) = 0.50$$

reqd ②

$$\frac{V_b - V_a}{20} + \frac{V_b - 5}{10} = -0.666 A$$

$$0.05 V_b - 0.05 V_a + 0.1 V_b = +0.5 - 0.666 A$$

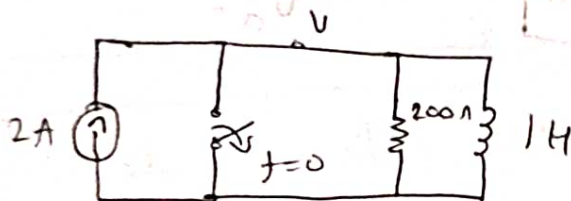
$$-0.05 V_a + 0.15 V_b = -0.166 A$$

$$V_a(0^+) = 1.90 V$$

$$V_b(0^+) = -0.475 V$$

13) In the n/w shown the switch K is opened at $t=0$ find

$$V, \frac{dV}{dt}, \frac{d^2V}{dt^2} \text{ at } t=0^+$$

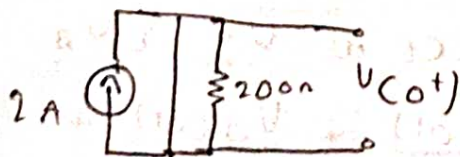


Soln: case (i) $t=0^-$

The $I = 0$ at inductor

$$I(0^-) = 0 A$$

case (ii) $t=0^+$

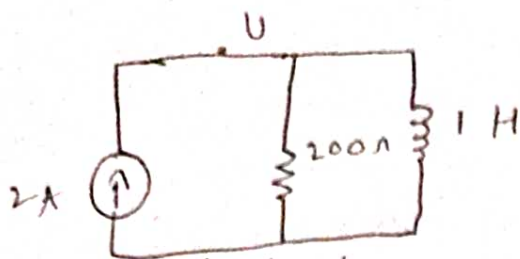


$$I(0^+) = I(0^-) = 0 A$$

$$V = I \times R$$

$$V(0^+) = 2 \times 200$$

$$V(0^+) = 400 \text{ V}$$



apply KVL

$$\frac{V}{200} + \frac{1}{L} \int V dt = 2 \quad \dots (1)$$

① 0.66 W s t t

$$\frac{1}{200} \frac{dV}{dt} + \frac{1}{L} V(0^+) = 0$$

$$\frac{dV(0^+)}{dt} = -\frac{400}{0.005}$$

$$\frac{dV}{dt} = -80000 \text{ V/s}$$

② 0. W s t t

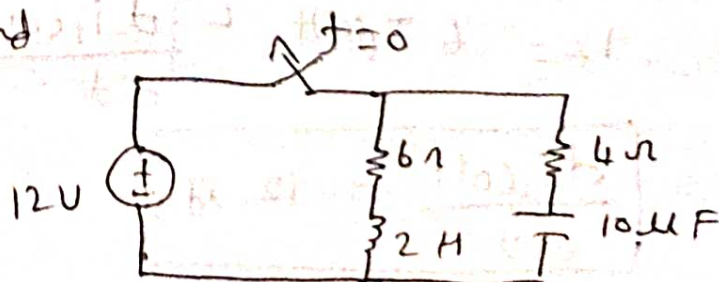
$$\frac{1}{200} \frac{d^2V}{dt^2} + \frac{1}{L} \frac{dV(0^+)}{dt} = 0$$

$$\frac{d^2V}{dt^2} = 16 \times 10^6 \text{ V/s}^2$$

14) For the circuit shown find

i) $i(t)$ and $V(t)$

ii) $\frac{di(t)}{dt}$ and $\frac{dV(t)}{dt}$



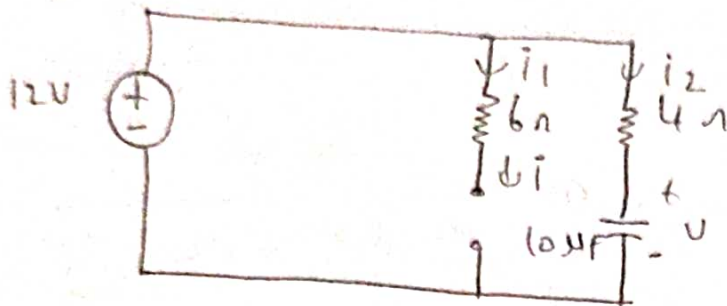
case i) at $t = 0^-$

$$i_1(0^-) = 0 \text{ A}$$

$$V(0^-) = 0 \text{ V}$$

$$i_2(0^-) = 0 \text{ A}$$

case ii) at $t = 0^+$



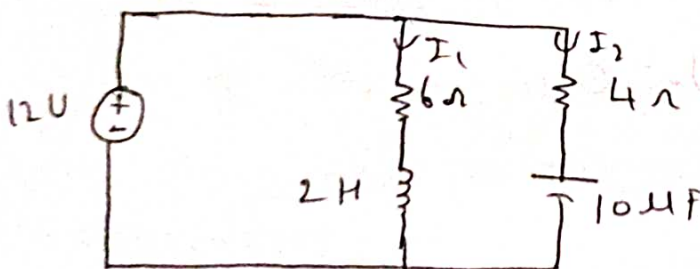
$$i_1(0^+) = 0 \text{ A}$$

$$i_2(0^+) = 3 \text{ A}$$

$$V(0^+) = \frac{4 \times 12}{4 + 6}$$

$$V(0^+) = 4.8 \text{ V}$$

case iii) at $t \rightarrow \infty$



apply KVL to loop ① & ②

$$12 = 6 I_1(t) + L \frac{d i_1(t)}{dt}$$

$$\frac{d i_1(t)}{dt} = 12 \times 1/5$$

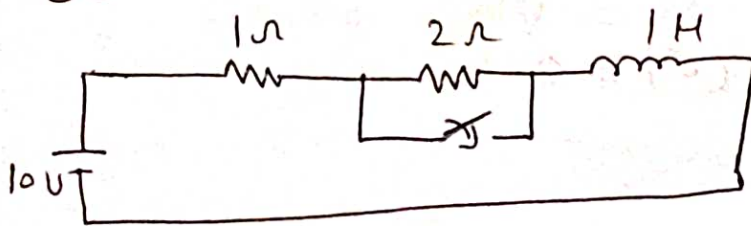
apply KVL to loop ②

$$12 = 4I_2 + \frac{1}{5} L \frac{di_2(t)}{dt}$$

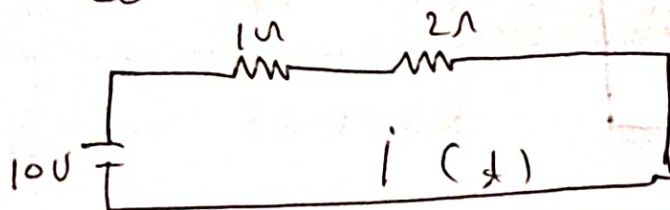
$$12 - 12 = \frac{di_2(t)}{dt}$$

$$\frac{di_2(t)}{dt} = 0 \text{ A/s}^2$$

15) For the ckt shown find $V_K(t)$, $\frac{dV_K(t)}{dt}$, $\frac{d^2V_K(t)}{dt^2}$

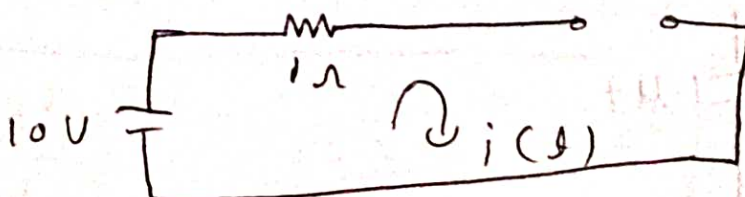


case i: at $t = 0^-$



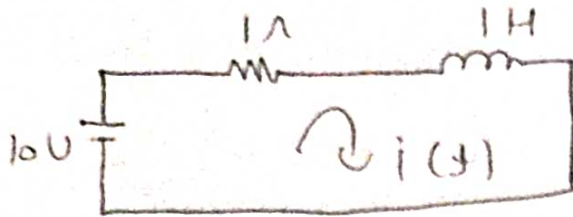
$$I(0^-) = \frac{10}{3} = 3.33 \text{ A}$$

case ii: at $t = 0^+$



$$I(0^+) = 0 \text{ A}$$

case iii) at $t > 0/\infty$



$$-i(t) + L \frac{dI}{dt}(0^+) = 10 \quad \text{--- ①}$$

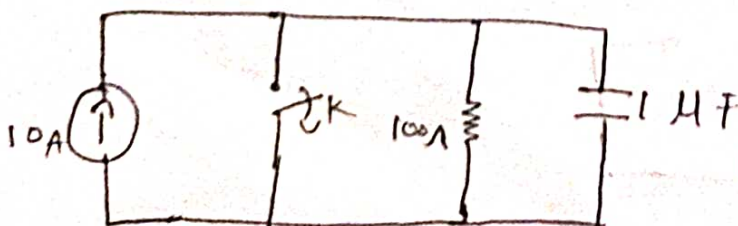
$$\boxed{\frac{dI}{dt}(0^+) = 10 \text{ A/s}}$$

$$\frac{dV}{dt}(0^+) = I \times R$$

$$- \frac{di(0^+)}{dt} - L \frac{d^2 I}{dt^2}(0^+) = 0$$

$$\boxed{\frac{d^2 I}{dt^2} = -10 \text{ A/s}^2}$$

16) In the circuit shown the switch K is opened at $t=0$ find the values of V , $\frac{dV}{dt}$, $\frac{d^2 V}{dt^2}$ at $t=0^+$



case i) at $t = 0^-$

$$i(0^-) = 0 \text{ A}$$

$$V(0^-) = 0 \text{ V}$$

case ii) at $t = 0^+$

$$i(0^+) = i(0^-) = 0 \text{ A}$$

$$V(0^+) = V(0^-) = 0 \text{ V}$$

case iii) apply KCL

$$\frac{V(t)}{100} + C \frac{dV}{dt} = 0 \quad \text{--- (1)}$$

$$\frac{dV}{dt} = -\frac{10}{1 \times 10^{-6}}$$

$$\boxed{\frac{dV}{dt} = -10 \text{ MV/s}}$$

ev @ diff w.r to t

$$\frac{1}{100} \frac{dV}{dt} + C \frac{d^2V}{dt^2} = 0$$

$$\boxed{\frac{d^2V}{dt^2} = -10 \times 10^{10} \text{ V/s}^2}$$